Retrieval-Augmented Generation for Recipe Question Answering: A Comparative Study of Open-Source LLMs

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Abstract

Retrieval-Augmented Generation (RAG) has emerged as a powerful paradigm for grounding large language models in domain-specific knowledge, reducing hallucinations and improving factual accuracy. In this paper, we present a comparative evaluation of three open-source instruction-tuned LLMs – Microsoft Phi-3 Medium (14B), Mistral-7B Instruct, and Meta Llama-3 8B – within a RAG system designed for recipe question answering. Using a corpus of 13 recipes from Simply Recipes, we built a FAISS-based vector index with chunked text and evaluated each model on 13 domain-specific questions. We assess performance using automated metrics (faithfulness, answer relevance, retrieval precision/recall) as well as qualitative analysis of accuracy, completeness, and helpfulness. Our results show that all three models achieve perfect retrieval recall, but Phi-3 delivers the highest answer relevance and factual correctness, while Llama-3 exhibits superior faithfulness to the retrieved context. Mistral performs competitively but makes occasional factual errors. We discuss trade-offs between model size, response conciseness, and reliability, offering practical insights for selecting LLMs in knowledge-intensive applications.

1. Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in text generation, question answering, and reasoning. However, their tendency to hallucinate – produce plausible but incorrect information – remains a critical limitation, especially in domains requiring up-to-date or proprietary knowledge. Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) addresses this by retrieving relevant documents from an external knowledge base and conditioning the LLM on that context. RAG has become a standard approach for building trustworthy question-answering systems in specialized domains (Guu et al., 2020; Asai et al., 2023).

The recent proliferation of open-source LLMs (e.g., Llama, Mistral, Phi) has democratized access to high-quality models. However, practitioners lack systematic comparisons of how these models perform when integrated into a RAG pipeline, particularly for task-oriented domains like cooking assistance. Recipe question answering presents unique challenges: answers are often short facts (e.g., time, temperature, ingredient quantity) but require exact adherence to the recipe text.

In this paper, we design a RAG system for answering questions about a curated collection of 13 recipes. We evaluate three prominent open-source instruction-tuned models – Phi-3 Medium (14B), Mistral-7B Instruct, and Llama-3 8B – using identical retrieval and prompting. Our contributions are:

· A reproducible RAG pipeline for recipe QA, including text chunking, embedding-based retrieval with FAISS, and a simple one-sentence prompting strategy.
· A set of 13 domain-specific questions with ground-truth answers derived directly from the recipes.
· Quantitative and qualitative evaluation across four metrics: faithfulness, answer relevance, retrieval precision, and recall.
· Analysis of each model's strengths and weaknesses, revealing trade-offs between conciseness, faithfulness, and factual accuracy.

Our findings indicate that while all models effectively use retrieved contexts, Phi-3 offers the best balance for a helpful cooking assistant, whereas Llama-3 excels at producing answers strictly tied to the source text.

2. Related Work

Retrieval-Augmented Generation. Lewis et al. (2020) introduced RAG, combining a dense passage retriever (DPR) with a BART generator. Subsequent work proposed REALM (Guu et al., 2020), which jointly pre-trains retrieval and generation, and Self-RAG (Asai et al., 2023), which adds reflection tokens to decide when to retrieve. For open-domain QA, RAG has been shown to outperform purely parametric models. Our work builds on these foundations but applies RAG to a narrow, recipe-specific corpus.

Open-Source LLMs. The release of Llama (Touvron et al., 2023) and its successors (Llama-2, Llama-3) has enabled reproducible research on large models. Mistral-7B (Jiang et al., 2023) offers a competitive 7B parameter model with sliding window attention. Phi-3 (Microsoft, 2024) is a 14B model trained with a focus on instruction following and reasoning. All three models are available in 4-bit quantized versions, making them viable on consumer GPUs.

Evaluation of RAG Systems. RAGAS (Es et al., 2023) provides a framework for automated evaluation of RAG pipelines using metrics like faithfulness, answer relevance, and context recall. Similarly, ARES (Saad-Falcon et al., 2023) offers lightweight evaluation. Our work adapts these metrics but implements them locally without external APIs, using embedding similarity and keyword overlap.

Recipe QA. RecipeQA (Yagcioglu et al., 2019) introduced a multimodal dataset for recipe comprehension, but it focuses on multiple-choice and cloze tasks rather than generative QA. More recent work (e.g., FoodChat) uses knowledge graphs for cooking dialogue. Our paper contributes a focused evaluation of LLMs in a RAG setting for recipe QA, grounded in real-world user questions.

3. System Design and Implementation

3.1 Knowledge Base

We collected 13 recipes from Simply Recipes (simplyrecipes.com), covering a diverse range of dishes: chicken wraps, scones, cauliflower nuggets, Cajun pasta, smoothie, carrot cake, lasagna, roast chicken, burgers, donuts, brownies, lemon cake, and blueberry muffins. Each recipe was saved as a plain text file (UTF-8) to avoid PDF extraction errors. The average recipe length was approximately 1,500 characters.

3.2 Chunking and Embedding

To enable efficient retrieval, we split each recipe into overlapping chunks using RecursiveCharacterTextSplitter (LangChain) with chunk size 800 characters and overlap 100 characters. This overlap ensures that relevant information spanning chunk boundaries is captured. After chunking, we obtained 47 chunks across all recipes. We used the all-MiniLM-L6-v2 sentence transformer (Reimers & Gurevych, 2019) to compute dense embeddings of dimension 384. This model is lightweight and well-suited for semantic similarity on short passages. All chunk embeddings were indexed using FAISS (Johnson et al., 2019) with IndexFlatL2 (Euclidean distance).

3.3 Retrieval

For a given user question, we compute its embedding and retrieve the top-6 most similar chunks (by L2 distance). These chunks constitute the contextual information provided to the LLM. No re-ranking or additional filtering is applied. The retrieved chunks are concatenated with \n\n---\n\n as a delimiter.

3.4 LLM Selection and Quantization

We selected three instruction-tuned models:

1. Meta Llama-3 8B (instruction variant)
2. Mistral-7B Instruct v0.2
3. Microsoft Phi-3 Medium (14B)

All models were loaded via the unsloth library, which implements 4-bit quantization (Dettmers et al., 2023) using bitsandbytes. We set max_seq_length=2048, dtype=torch.bfloat16, and device_map="auto" for automatic GPU/CPU offloading. This configuration fits within the 16 GB VRAM of a Google Colab T4 GPU for each model individually. We restarted the runtime between models to avoid memory conflicts.

3.5 Prompting and Generation

We used a simple, direct prompt to minimize variance across models:

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Text: {context}
Question: {query}
Answer (one sentence):
```

We set max_new_tokens=150, temperature=0.1, and do_sample=True to encourage deterministic, factual outputs. A stop_string=["?"] was used to cut off the response after a question mark, which helped maintain conciseness. After generation, we performed a cleanup step: extracting the text after "Answer (one sentence):", splitting on "Question:" or double newlines, and ensuring the answer ends with a period. Encoding fixes were applied for degree symbols.

3.6 Domain-Specific Questions

We manually crafted 13 questions, each answerable from a single recipe. The questions cover various aspects: storage duration, technique purpose, ingredient substitutions, nutritional facts, cooking temperatures, and times. The full list appears in Appendix A (Table 4). For each question, we also recorded the ground-truth answer and the source recipe ID.

4. Evaluation Methodology

4.1 Automated Metrics

We defined four metrics to evaluate both retrieval and generation:

- **Faithfulness:** For each answer, we computed the proportion of content words (excluding stopwords) that appear in any of the retrieved chunks. This provides a simple measure of how grounded the answer is in the retrieved context.
- **Answer Relevance:** We used cosine similarity between the question embedding and the answer embedding (both generated by all-MiniLM-L6-v2). Higher similarity indicates a more on-topic answer.
- **Retrieval Precision:** Among the recipes listed in the sources field (derived from retrieved chunks), we computed the percentage that matched the ground-truth recipe for that question.
- **Retrieval Recall:** Binary indicator of whether the ground-truth recipe appeared at least once in the retrieved sources.

All metrics were computed using the same embedding model to ensure consistency.

4.2 Qualitative Evaluation

We performed a human evaluation on a subset of 10 questions. Two independent annotators rated each model's answer on a Likert scale (1–3) for:

- Accuracy: Factual correctness relative to the recipe.
- Completeness: Whether the answer includes all necessary details.
- Helpfulness: Clarity, conciseness, and usefulness for a home cook.

Inter-rater agreement was high (Cohen's $\kappa = 0.86$). The final score for each model is the average of the two annotators.

4.3 Experimental Setup

All experiments were run on Google Colab with a single NVIDIA T4 GPU (16 GB VRAM) and a CPU with 13 GB RAM. For each model, we executed the same evaluation loop: for each of the 13 questions, we retrieved chunks, generated an answer, and recorded metrics. The process was repeated after restarting the runtime for each model. The total runtime per model was approximately 5 minutes.

5. Results

5.1 Quantitative Results

Table 1 presents the average metrics for each model across all 13 questions. All models achieved perfect retrieval recall (1.0), indicating that the correct recipe was always among the top-6 retrieved chunks. Retrieval precision was moderate (0.42–0.44), meaning that about half of the retrieved recipes were irrelevant; however, the presence of the correct recipe sufficed for answering.

Model	Faithfulness	Answer Relevance	Retrieval Precision	Retrieval Recall
Phi-3	0.757	0.828	0.420	1.000
Mistral	0.770	0.807	0.423	1.000
Llama-3	0.792	0.761	0.419	1.000

Phi-3 achieved the highest answer relevance (0.828), indicating that its responses were semantically closest to the intended answers. Llama-3 led in faithfulness (0.792), meaning its answers were most tightly grounded in the retrieved chunks. Mistral's faithfulness was slightly lower (0.770) but comparable.

5.2 Qualitative Results

Table 2 shows the averaged human ratings (1–3) for accuracy, completeness, and helpfulness based on 10 questions.

Model	Accuracy	Completeness	Helpfulness
Phi-3	2.95	2.85	2.90
Mistral	2.65	2.70	2.75
Llama-3	2.80	2.45	2.60

Phi-3 was nearly perfect in accuracy, with only one minor issue (a slightly off phrase in the scones answer). Llama-3 was also accurate but often too brief, sacrificing completeness. Mistral made one notable factual error: it stated that the Easy Roast Chicken contains 37g of protein per serving, while the correct value is 60g. This error lowered its accuracy score.

5.3 Example Answers

Table 3 shows representative answers for two questions. Llama-3's answer is factually correct but lacks a complete sentence, reducing helpfulness.

Question	Model	Answer
What is the protein content per serving in the Easy Roast Chicken?	Phi-3	The protein content per serving in the Easy Roast Chicken is 60g.
	Mistral	The protein content per serving in the Easy Roast Chicken is 37g.
	Llama-3	The protein content per serving in the Easy Roast Chicken is 60g.
What ingredient is used to make the breading mixture 'cheesy' without cheese?	Phi-3	Nutritional yeast is used to make the breading mixture 'cheesy' without cheese.
	Mistral	Nutritional yeast is used to make the breading mixture 'cheesy' without cheese.
	Llama-3	Nutritional yeast.

6. Discussion

6.1 Strengths and Weaknesses of Each Model

Phi-3 (14B)

· *Strengths:* Produced complete, well-formed sentences; consistently accurate; high answer relevance. Appears to understand implicit instructions (e.g., "one sentence") reliably.
· *Weaknesses:* Slightly lower faithfulness than Llama-3; occasionally added extraneous words. Larger model size (14B) may be overkill for simple QA.

Mistral-7B

· *Strengths:* Good balance of faithfulness and relevance; response quality similar to Phi-3. Faster inference than Phi-3 due to smaller size.
· *Weaknesses:* Made a factual error (protein content) that was not present in the retrieved context – a classic hallucination. This suggests that despite retrieval, Mistral sometimes overrides context with parametric knowledge.

Llama-3 8B

· *Strengths:* Highest faithfulness – its answers were almost entirely extractive from the retrieved chunks. Very concise, which can be beneficial for users who want quick facts.
· *Weaknesses:* Over-conciseness reduced completeness and helpfulness (e.g., "Nutritional yeast." without context). Lower answer relevance due to brevity.

6.2 Differences in Reasoning and Handling of Implicit Information

All models correctly answered most questions that required simple fact extraction. For questions requiring inference, such as "Why do you press down on the patty?", all three provided reasonable explanations. However, Llama-3 sometimes gave a one-word reason while Phi-3 and Mistral elaborated slightly. None of the models exhibited sophisticated multi-step reasoning, but the task did not demand it.

6.3 Effect of Retrieval on Accuracy

The perfect recall demonstrates that our retrieval component is effective for this small, well-structured corpus. However, retrieval precision was only ~0.42, meaning that the LLM often received irrelevant chunks. Despite this, models were generally able to ignore noise and extract the correct information – a testament to their instruction-following abilities. Llama-3's high faithfulness suggests it is particularly adept at ignoring irrelevant context, while Mistral's one error may indicate it was influenced by incorrect information.

6.4 Trade-offs: Model Size vs. Performance

Phi-3 (14B) performed best overall but required more GPU memory. Mistral (7B) and Llama-3 (8B) are smaller and faster, with Llama-3 providing the best faithfulness at lower cost. For applications where concise, extractive answers are acceptable, Llama-3 is a strong choice. For a user-facing assistant where natural sentences matter, Phi-3 or Mistral are preferable.

6.5 Limitations

Our study has several limitations. The small corpus (13 recipes) and narrow question set may not generalize to larger-scale or more diverse domains. The faithfulness metric based on word overlap is crude and may penalize paraphrasing. We did not test different chunk sizes, retrieval strategies (e.g., MMR), or prompt designs. Additionally, we used a single judge model (the same Llama-3 for all), though we mitigated this by incorporating human evaluation for qualitative metrics.

7. Conclusion and Future Work

We presented a comparative evaluation of three open-source LLMs – Phi-3, Mistral, and Llama-3 – within a RAG system for recipe question answering. Our results show that all three models effectively leverage retrieved contexts, with Llama-3 achieving the highest faithfulness and Phi-3 the best overall user-facing quality. Mistral performed competitively but exhibited one hallucination. The choice of model should therefore depend on the application's priority: fidelity to source text (Llama-3), natural language fluency (Phi-3), or a balance with smaller footprint (Mistral).

Future work should extend the evaluation to a larger recipe corpus and include more challenging questions requiring multi-step reasoning across multiple recipes. Additionally, we plan to explore prompt optimization techniques, dynamic chunk retrieval, and the use of hybrid search (dense + sparse) to improve precision. Finally, deploying the system as a real-time cooking assistant and collecting user feedback would provide valuable qualitative insights.

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Appendix A: Domain-Specific Questions and Ground Truth

Table 4: Questions with ground truth answers and source recipe

#	Question	Answer	Source Recipe
1	How many days can the chicken Caesar wraps be refrigerated?	Up to 2 days	Chicken Caesar Wrap
2	What is the purpose of chilling the scones before baking?	Solidify the butter, create flaky texture	Chocolate Chip Scones
3	What ingredient is used to make the breading mixture 'cheesy' without cheese?	Nutritional yeast	Crispy Breaded Cauliflower Nuggets
4	How many calories are in one serving of the 3-Ingredient Cajun Alfredo Pasta?	504	3-Ingredient Cajun Alfredo Pasta
5	What is the protein content per serving in the Easy Roast Chicken?	60g	Easy Roast Chicken
6	How long should the cake cool before frosting?	30 minutes	Divorce Carrot Cake
7	What is the purpose of spraying aluminum foil before covering the lasagna?	Prevent sticking	Easy Lasagna Primavera
8	At what internal temperature does the USDA recommend the chicken breast to be fully cooked?	165°F	Easy Roast Chicken
9	Why do you press down on each patty with a spatula for 10 seconds after placing it in the skillet?	Encourage browning	How To Make a Burger
10	What is the recommended oil temperature for frying the donuts?	350°F	How To Make Donuts
11	How can you prolong the moistness of the brownies after cooling?	Leave them uncut	Malted Chocolate Brownies
12	On which day does the cake have the best flavor according to the recipe?	Second day	One-Bowl Lemon Snack Cake
13	What two oven temperatures are used to bake the muffins, and for how long at each?	425°F for 5 min, then 375°F for 13–16 min	Quick Blueberry Muffins